



Effects of Resistance Training to Muscle Failure on Acute Fatigue: A Systematic Review and Meta-Analysis

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Abstract

Background Proper design of resistance training (RT) variables is a key factor to reach the maximum potential of neuromuscular adaptations. Among those variables, the use of RT performed to failure (RTF) may lead to a different magnitude of acute fatigue compared with RT not performed to failure (RTNF). The fatigue response could interfere with acute adaptive changes, in turn regulating long-term adaptations. Considering that the level of fatigue affects long-term adaptations, it is important to determine how fatigue is affected by RTF versus RTNF.

Objective The aim of this systematic review and meta-analysis was to compare the effects of RTF versus RTNF on acute fatigue.

Methods The search was conducted in January 2021 in seven databases. Only studies with a crossover design that investigated the acute biomechanical properties (vertical jump height, velocity of movement, power output, or isometric strength), metabolic response (lactate or ammonia concentration), muscle damage (creatinine kinase activity), and rating of perceived exertion (RPE) were selected. The data (mean $\pm$ standard deviation and sample size) were extracted from the included studies and were either converted into the standardized mean difference (SMD) or maintained in the raw mean difference (RMD) when the studies reported the results in the same scale. Random-effects meta-analyses were performed.

Results Twenty studies were included in the systematic review and 12 were included in the meta-analysis. The main meta-analyses indicated greater decrease of biomechanical properties for RTF compared with RTNF (SMD -0.96 , 95% confidence interval [CI] -1.43 to -0.49 , $p < 0.001$). Furthermore, there was a larger increase in metabolic response (RMD $4.48 \text{ mmol}\cdot\text{L}^{-1}$, 95% CI 3.19 – 5.78 , $p < 0.001$), muscle damage (SMD 0.76 , 95% CI 0.31 – 1.21 , $p = 0.001$), and RPE (SMD 1.93 , 95% CI 0.87 – 3.00 , $p < 0.001$) for RTF compared with RTNF. Further exploratory subgroup analyses showed that training status ($p = 0.92$), timepoint ($p = 0.89$), load ($p = 0.10$), and volume ($p = 0.12$) did not affect biomechanical properties; however, greater loss in the movement velocity test occurred on upper limbs compared with lower limbs ($p < 0.001$). Blood ammonia concentration was greater after RTF than RTNF (RMD $44.66 \text{ }\mu\text{mol}\cdot\text{L}^{-1}$, 95% CI 32.27 – 57.05 , $p < 0.001$), as was 48 h post-exercise blood creatine kinase activity (SMD 0.86 , 95% CI 0.33 – 1.42 , $p = 0.002$). Furthermore, although there was considerable heterogeneity in the overall analysis ($I^2 = 83.72\%$; $p < 0.01$), a significant difference in RPE after RTF compared with RTNF was only found for studies that did not equalize training volumes.

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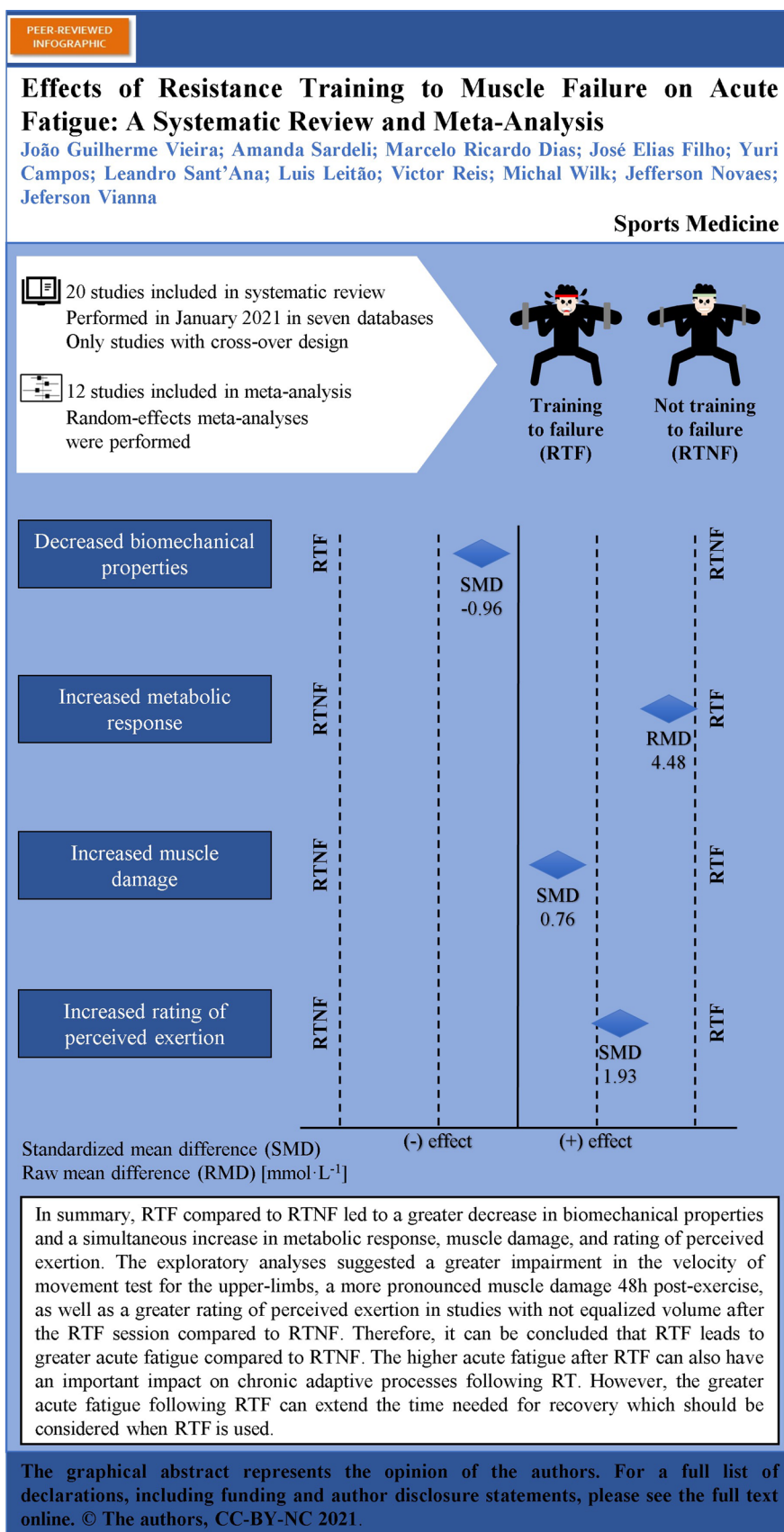
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Conclusions In summary, RTF compared with RTNF led to a greater decrease in biomechanical properties and a simultaneous increase in metabolic response, higher muscle damage, and RPE. The exploratory analyses suggested a greater impairment in the velocity of movement test for the upper limbs, more pronounced muscle damage 48 h post-exercise, and a greater RPE in studies with non-equalized volume after the RTF session compared with RTNF. Therefore, it can be concluded that RTF leads to greater acute fatigue compared with RTNF. The higher acute fatigue after RTF can also have an important impact on chronic adaptive processes following RT; however, the greater acute fatigue following RTF can extend the time needed for recovery, which should be considered when RTF is used.

Protocol Registration The original protocol was prospectively registered (CRD42020192336) in the International Prospective Register of Systematic Reviews (PROSPERO).

Graphical abstract



Key Points

Resistance training (RT) performed to failure (RTF) led to greater decline of biomechanical properties, higher metabolic response, and higher rating of perceived exertion compared with RT not performed to failure (RTNF).

RTF led to higher movement velocity decline during upper-limb exercise compared with lower-limb exercise.

RTF led to higher muscle damage than RTNF at 48 h of recovery.

1 Introduction

Resistance training (RT) increases maximal strength, power output, speed, strength endurance, balance, coordination, and body mass in athletes and non-athletes [1]. Proper design of training variables, such as the type of muscle contraction, external load used, number of repetitions performed, number of sets, length of the rest interval, velocity of movement, type and sequence of exercises, training frequency, and the level of effort, influences the magnitude of acute responses and, consequently, long-term adaptations [2–5]. Among these variables, the effects of RT performed to failure (RTF), which refers to the use of the maximum number of repetitions in each set, have been widely explored [6–9].

The greater effort applied during RTF may require the recruitment of motor units with a higher excitability threshold [10, 11] and a greater increase in metabolic response and muscle damage compared with RT not performed to failure (RTNF) [7–9, 12]. In this way, greater improvements in muscular adaptations would be expected after RTF [13, 14]. Nonetheless, recent meta-analyses have not confirmed larger improvements in maximal strength, muscle hypertrophy, or power output after RTF compared with RTNF [15–18].

Although the importance of RTF in long-term adaptations remains elusive, it may play a role in the magnitude of acute fatigue. Fatigue is defined as a response that is less than the expected or anticipated muscular contractile response to a given stimulus [19], reduced maximal strength and power output due to a decreased ability of the central nervous system to elicit activation of motor units, biochemical changes at the neuromuscular junction, and decreased muscle enzyme activity through the accumulation of hydrogen ions in the muscle and bloodstream [19–22]. Although fatigue provides one of the necessary stimuli for the occurrence of the general adaptation syndrome [23], inappropriate fatigue

management may increase the risk of injury, overtraining, and psychological burnout [24]. For example, RTF protocols that lead to greater fatigue response than RTNF also cause greater anxiety and tiredness [25], as well as a decrease in vertical jump height [7, 8, 26], velocity of movement [7, 8, 26], power output [12, 27, 28], and isometric strength [29–31]. These impairments may persist for more than 24 h after RT sessions.

Physiologically, RTF breaks cellular homeostasis by depleting phosphocreatine storages, reducing adenosine triphosphate and the total muscle adenine nucleotides pool [27], leading to tissue damage [7, 8, 32], an increase in metabolic fatigue markers [12, 27, 33–35], and a reduction in neuromuscular performance [9, 12, 27]. On the other hand, some studies observed a small or non-significant difference in vertical jump height, power output, isometric strength, central fatigue, or rating of perceived exertion (RPE) between RTF and RTNF [6, 36]. These divergent findings could be caused by different characteristics of populations, RT protocols, timepoints, methods of assessments, and type of fatigue response. Regarding the characteristics of the population, it is already known that significant differences in fatigue response exist between men and women [29, 30] and young and older participants [37]. Additionally, the majority of studies comparing fatigue response between RTF and RTNF have considered relatively small sample sizes [7, 26, 27, 29].

When assessing the differences in acute fatigue between RTF and RTNF, it is of fundamental importance to assign an appropriate weight to each study as part of an integrative analysis. Therefore, the aim of the present study was to compare RTF versus RTNF on acute fatigue in adult males, considering each type of fatigue response (biomechanical properties, metabolic responses, muscle damage, and RPE) at different timepoints after the RT session. The pooled meta-analysis will clarify the understanding of overall fatigue responses within a comparable population. Furthermore, the possible causes of the remaining heterogeneity will be further investigated. We hypothesized that RTF would cause a higher decline in biomechanical properties and a greater increase in metabolic response, muscle damage, and RPE. The results of this meta-analysis can be useful in understanding the RT effects on fatigue and could help in optimization of RT programs by coaches and practitioners.

2 Methods

A systematic review and meta-analysis of the literature was performed according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [38].

2.1 Protocol and Registration

The original protocol was registered in the International Prospective Register of Systematic Reviews (PROSPERO) on 18 July 2020 (registration number: CRD42020192336).

2.2 Eligibility Criteria

Five PICOS criteria [39] were used: (1) a population of healthy men aged between 18 and 40 years, trained or untrained in RT, without a history of bone, muscle, or articular injury; (2) RT sessions performed to failure; (3) compared with RT sessions not performed to failure; (4) assessing, as outcomes, acute fatigue for biomechanical properties (vertical jump height, velocity of movement, power output, and isometric strength), metabolic response (lactate and ammonia concentration), muscle damage (creatine kinase activity), and RPE; (5) studies with a randomized controlled design, counterbalanced crossover or repeated measures design RTF and RTNF.

The age range as an inclusion criterion was determined in accordance with previous studies [40, 41]. In addition, after 40 years of age, significant changes are observed, including in the resting concentration of hormones (e.g., testosterone), which have a significant impact on muscle adaptation after RT [42, 43]. Finally, we considered RTF studies describing their RT protocols as concentric muscle failure, maximum number of repetitions, and maximum effort with no repetition in reserve. For RTNF, the absence of these terms indicated the absence of RTF.

2.3 Selection Criteria

The inclusion criteria adopted for the study selection were (1) original studies; (2) RT-based intervention; and (3) studies assessing at least one of the outcomes of interest. Exclusion criteria were (1) duplicated studies; (2) studies not written in the English language; (3) studies not isolating RT effects such as those combining RT with other types of training (aerobic, flexibility, etc.); and (4) studies testing RT effects on populations with specific health conditions (children, pregnant women, elderly individuals, people with lower back pain, coronary artery disease, osteoarthritis, hypertension, diabetes or obesity).

2.4 Information Sources

The studies were retrieved from an electronic database search and from a comprehensive sweeping in the reference list of the included studies. A search of the following databases was conducted on January 2021: Cumulative Index to Nursing and Allied Health (CINAHL), Cochrane Library,

EMBASE, PubMed, Scopus, SPORTDiscus, and Web of Science.

2.5 Search Strategy

A previous study [15] supported the selection of adequate descriptors for the search strategy. The search strategy combined the following descriptors and Boolean operators (AND/OR/NOT): (“resistance training” OR “resistance exercise” OR “strength training” OR “strength exercise” OR “weight training” OR “weight exercise” OR “weightlifting” OR “weight-lifting” OR “weight lifting”) AND (“repetition failure” OR “repetition to failure” OR “repetitions to failure” OR “muscle failure” OR “muscular failure” OR “momentary failure” OR “failure” OR “failure training” OR “nonfailure” OR “non-failure” OR “not to failure” OR “volitional interruption”) NOT (“review” OR “blood flow restriction” OR “heart failure” OR “supplement” OR “obesity”). The equivalent search syntax for each database is described in the electronic supplementary material (ESM).

2.6 Study Selection

The studies retrieved in each database were sent to the EndNote X9 software (Clarivate Analytics, Philadelphia, PA, USA), and the duplicate studies were automatically and manually removed. Titles and abstracts were assessed according to the eligibility criteria by two independent researchers (JGV and MRD). Any conflicts were decided by a third reviewer (JEF). The researchers were not blinded to authors, institutions, or journals. The abstracts lacking decisive information were selected for full-text inspection. When information was absent or incomplete, the authors of the studies were contacted by email.

2.7 Data Collection Process

Two independent reviewers (JGV and MRD) extracted the data from the full-texts, using a standardized and previously structured protocol. The data collected included the participant characteristics (sample size, age, height, body mass, and training status) and training protocols (study design, resistance exercise[s], exercise protocols, velocity of movement, training volume, and outcome measure[s]). When quantifiable data were not shown, data were extracted from graphs (JGV) using WebPlotDigitizer software, version 4.2 (San Francisco, CA, USA). The data extracted by both reviewers were compared and any divergences were decided by both reviewers as well as a third reviewer (JV).

2.8 Risk of Bias in the Included Studies

Risk of bias was assessed by two independent authors (JGV and JEF) using the Cochrane Collaboration's tool for assessing the risk of bias in randomized trials [44]. Selection bias (random sequence generation and allocation concealment), performance bias (blinding of participants and researchers), detection bias (blinding of outcome assessment), attrition bias (incomplete outcome data), reporting bias (selective reporting), and other bias (anything else, ideally prespecified) were evaluated.

2.9 Statistical Analysis

The meta-analyses were performed using the Comprehensive Meta-Analysis (CMA) software, version 3.3.070 (Biostat Inc., Englewood, NJ, USA), with the level of significance set at $p < 0.05$. We performed four main meta-analyses, one for each outcome (biomechanical properties, metabolic response, muscle damage, and RPE). The effect size (ES) was calculated based on the difference of variations (pre/post) between RTF and RTNF (A), or difference of post-exercise assessments between RTF and RTNF (B) when the studies did not show pre-values. Biomechanical properties and metabolic responses were analyzed based on the A and B designs, while muscle damage was analyzed based on only the A design and RPE was analyzed based on only the B design. When the variables were presented in the same units in all studies, we calculated the raw/absolute effect (metabolic response). However, when the variables were not presented in the same units in all studies, we calculated the standardized effect based on each study's standard deviation (biomechanical properties, muscle damage, and RPE). A random-effects model was used in all meta-analyses [45]. The magnitude of the ES was assumed as trivial (< 0.2), small (0.20–0.49), moderate (0.50–0.79), or large (> 0.80) based on the classifications set out by Cohen [46]. Heterogeneity between the studies was quantified through the I^2 statistic. The results of this test indicate the percentage of inconsistency between studies. The inconsistency was classified according to the following scale: low ($< 25\%$), moderate (25–75%), and high ($> 75\%$) [47]. Publication bias was analyzed using Egger's test and a p -value of 5% ($p < 0.05$) was considered significant [48].

To avoid sample overlapping and enable consistent analyses, we selected just one protocol from each study for the main analyses, according to the most frequent type of outcome, load of the protocol, and timepoint assessed. In this way, regarding the type of outcome within biomechanical properties meta-analysis, the height of countermovement jump (CMJ) was selected, and within metabolic response meta-analysis, lactate concentration was chosen. Considering the different protocols, we gave preference for the

75% 1RM or 10RM loads [8, 9, 49]. The timepoint selected for biomechanical properties and metabolic response meta-analyses was immediately post-exercise, while for RPE meta-analysis the accepted timepoint ranged from immediately post-exercise to the 30th min of recovery, and for the muscle damage meta-analysis the preferred time point was 48 h post-exercise.

Within biomechanical properties and muscle damage meta-analyses, subgroup analyses were performed to compare the RTF effects on each timepoint (immediately post-exercise, and 6, 24, and 48 h post-exercise). Regarding muscle damage analysis, we did not include the 6 h timepoint, reported in one study [7], as creatine kinase activity increases approximately 100% only within 8 h after RT and its peak levels are observed between 24 and 96 h after exercise [50]. Training volume (equalized and not equalized) was assessed for its influence on biomechanical properties, metabolic response, and RPE. Additionally, within biomechanical properties meta-analysis, we also tested subgroups of training status (trained and untrained), type of test (CMJ, maximal voluntary isometric contraction [MVIC], velocity of movement against V_1 -load for bench press [velocity BP], and velocity of movement against V_1 -load for squat [velocity SQ]), and RT load (70, 75, 80, 85, 90% 1RM). The difference between RTF and RTNF on metabolic response was also tested within a group of studies assessing ammonia as the metabolic marker, instead of lactate. The Q-test was used to identify differences between categories of subgroups, considered significant when the p -value was < 0.05 . Finally, some studies lacking information have to be removed from specific subgroup analysis in case they could not be surely addressed to one or another subgroup.

2.10 Certainty of Evidence

The certainty of evidence was assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach [51]. The GRADE approach suggests the classification of randomized controlled trials initially as high-quality studies (score 4), that go through the specific risk-of-bias assessments to identify whether their scores need to be reduced to moderate, low, or very low. The following topics were assessed: (1) quality of the original studies; (2) inconsistency of the results (heterogeneity); (3) indirect evidence; (4) imprecision; and (5) publication bias. One point was deducted from the quality of the original studies when 50% of the studies in a determined meta-analysis had more than one item assessed as high risk. For inconsistency, we deducted a point if statistical heterogeneity was found. The risk of indirect evidence was assessed considering three factors: (1) when the participants differed from the population of interest; (2) when the interventions differed from the specific desired intervention; and (3) when

substitute outcomes were used instead of the relevant outcomes. Imprecision was assessed based on a total sample size of fewer than 100 participants. Regarding publication bias, we deducted one point from the quality score for studies with a significant risk on Egger's test ($p < 0.05$).

3 Results

3.1 Study Selection

The flow diagram of the literature search is presented in Fig. 1. Among the 4439 studies retrieved in the database searches, 12 were selected for at least one of the four meta-analyses.

3.2 Study Characteristics

All included studies followed a crossover design with a recovery period (range 2–60 days) between intervention sessions, except for the study by Vasquez et al. [28] (10 min between each condition). Overall, 179 participants were analyzed, of whom 116 (64.8%) were trained and 46 (25.7%) untrained. Two studies (9.5%) did not clearly report the training status of participants and were excluded from this specific subgroup analysis [33]. In general, the majority of exercise protocols were composed of one to two multi-articular exercises, while some protocols were composed of one to two mono-articular exercises in a variety of equipment. The load of the RTF protocols was equal to or above 40% 1RM (range 40–90%), while some defined their load based on 3, 4, 6, 8, 10, and 12 maximum repetitions; when studies presented more than one RTNF protocol, the most similar to RTF was selected [7, 9, 28, 32, 35, 52]. Details of

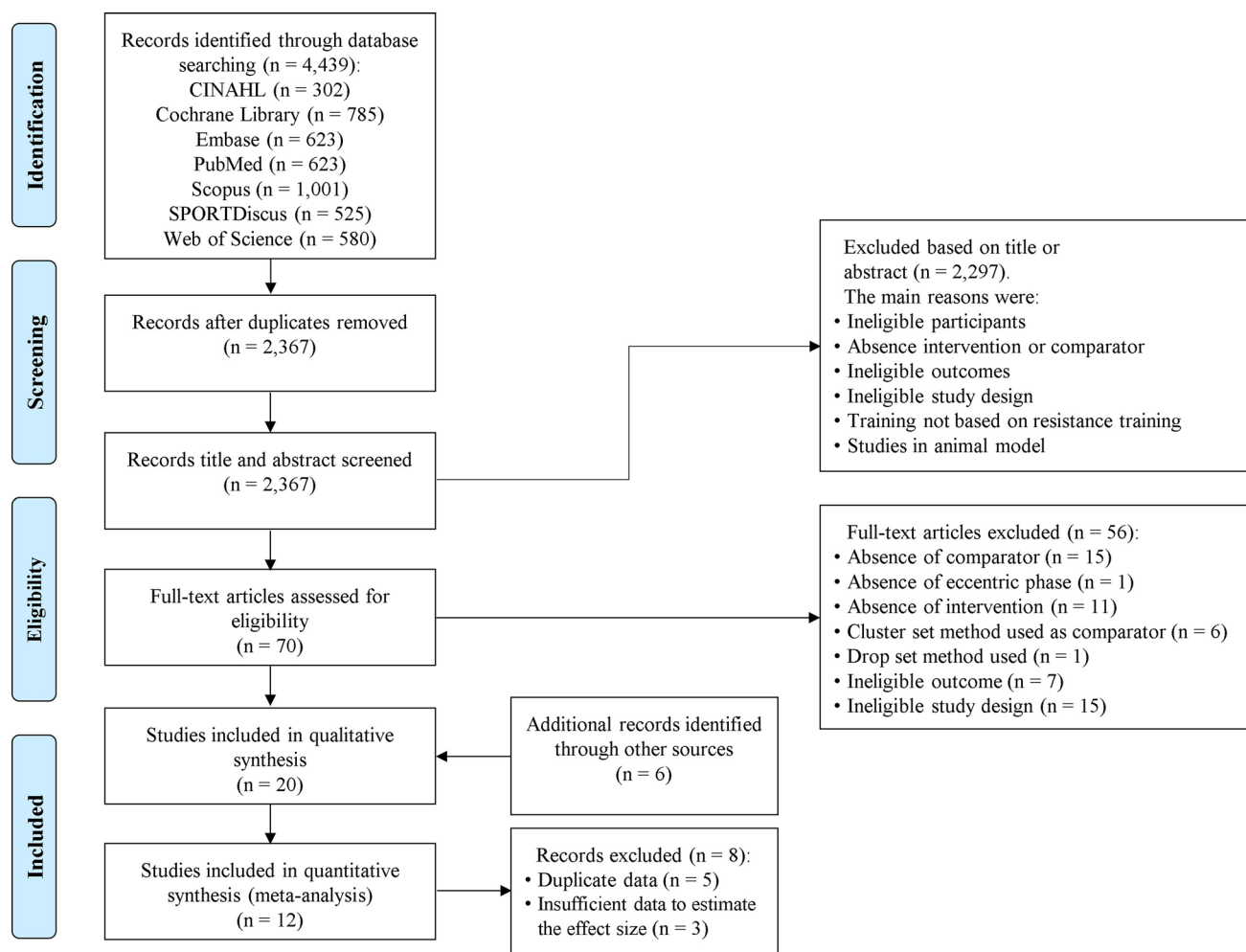


Fig. 1 PRISMA flow diagram of study selection. *PRISMA* Preferred Reporting Items for Systematic Reviews and Meta-Analyses

the characteristics of the participants and the included studies are presented in Tables 1 and 2.

3.3 Risk of Bias in the Included Studies

Figure 2 shows the risk of bias for each study. Most studies had an unclear risk of bias for random sequence generation and allocation concealment, however one study reported the use of random sequence generation [36] and three studies had a high risk of bias for random sequence generation and allocation concealment [29–31]. All studies had an unclear risk of bias for blinding of participants and personnel, blinding of outcome assessment, and selective reporting. All studies had a low risk of bias for incomplete outcome data. Six studies had a high risk of bias for other bias [25, 28, 31, 33, 36, 53].

3.4 Main Outcomes

The main meta-analysis of the biomechanical properties showed that the reduction was higher for RTF compared with RTNF (standardized mean difference [SMD] -0.96 , 95% confidence interval [CI] -1.43 to -0.49 , $p < 0.001$) (Fig. 3a). Table 3 shows subgroup analysis, in which training status ($p = 0.92$), timepoint ($p = 0.89$), load ($p = 0.10$),

and volume ($p = 0.12$) did not affect the difference between RTF and RTNF on biomechanical properties of fatigue. There was higher reduction in biomechanical properties with RTF compared with RTNF within the velocity BP test (SMD -2.47 , 95% CI -3.35 to -2.13 , $p < 0.001$) than in the other tests within this subgroup ($p < 0.001$). RTF led to a higher significant reduction in biomechanical properties compared with RTNF along all timepoints assessed (immediately and 6, 24, and 48 h after the RT session).

Lactate (raw mean difference [RMD] $4.48 \text{ mmol}\cdot\text{L}^{-1}$, 95% CI 3.19 – 5.78 , $p < 0.001$) [Fig. 3b] and ammonia concentration (RMD $44.66 \text{ }\mu\text{mol}\cdot\text{L}^{-1}$, 95% CI 32.27 – 57.05 , $p < 0.001$) were significantly higher after RTF compared with RTNF. Subgroup analysis showed that training volume did not influence lactate concentrations ($p = 0.77$). The main muscle damage meta-analysis included one study assessing CK activity at 24 h post exercise [49]; the CK activity increased significantly more in RTF compared with RTNF (SMD 0.76 , 95% CI 0.31 – 1.21 , $p = 0.001$) [Fig. 3c]. Subgroup analysis showed that after only 48 h, creatine kinase activity was higher in RTF (SMD 0.86 , 95% CI 0.33 – 1.42 , $p = 0.002$), but no difference was observed between RTF and RTNF immediately post and 24 h post exercise (Table 3). Finally, RPE was significantly higher in RTF than RTNF (SMD 1.93 , 95% CI 0.87 – 3.00 , $p < 0.001$) (Fig. 3d). Subgroup analysis

Table 1 Participant characteristics

Study	Participants ($n = 218$)	Age (years)	Height (cm or m)	Body mass (kg)	Training status
Arent et al. [25]	15	22.0 ± 0.7	180.8 ± 1.9	77.8 ± 2.9	Untrained
Fonseca et al. [36]	22	21.4 ± 2.3	Not reported	78.1 ± 6.7	Trained
González-Badillo et al. [26]	9	23.3 ± 3.9	1.75 ± 0.03	75.3 ± 9.2	Trained
González-Hernández et al. [6]	12	23.6 ± 1.5	178.2 ± 6.6	76.5 ± 9.9	Trained
Gorostiaga et al. [12]	13	34.4 ± 5.4	177.4 ± 6.0	74.1 ± 6.3	Untrained
Gorostiaga et al. [27]	6	34.0 ± 6.0	179.0 ± 5.0	74.5 ± 7.2	Untrained
Gorostiaga et al. [34]	6	34.0 ± 6.0	179.0 ± 5.0	74.5 ± 7.2	Untrained
Hiscock et al. [52]	10	26.3 ± 8.4	181.3 ± 5.6	78.1 ± 9.1	Trained
Linnamo et al. [29]	8	27.1 ± 2.0	181.3 ± 3.1	74.4 ± 9.1	Untrained
Linnamo et al. [30]	8	27.1 ± 2.0	181.3 ± 3.1	74.4 ± 9.1	Untrained
Martorelli et al. [49]	12	24.2 ± 4.4	177.1 ± 3.3	82.1 ± 6.4	Trained
McGuigan et al. [53]	8	21.6 ± 1.2	1.80 ± 0.10	86.6 ± 11.0	Unclear
Morán-Navarro et al. [7]	10	21.5 ± 4.0	175.2 ± 7.2	72.4 ± 8.4	Trained
Pareja-Blanco et al. [8]	10	22.1 ± 3.5	1.75 ± 0.07	73.5 ± 10.7	Trained
Pareja-Blanco et al. [32]	10	20.6 ± 2.7	1.75 ± 0.10	71.7 ± 12.5	Untrained
Pareja-Blanco et al. [58]	10	23.6 ± 3.7	1.75 ± 0.03	75.0 ± 8.7	Trained
Raastad et al. [33]	9	26.9 ± 4.2	Not reported	81.4 ± 9.6	Unclear
Sánchez-Medina and González-Badillo [9]	18	25.6 ± 3.4	176.6 ± 7.5	75.9 ± 9.1	Trained
Shibata et al. [31]	10	20.5 ± 1.1	174.0 ± 3.8	65.7 ± 4.8	Trained
Vasquez et al. [28]	12	21.9 ± 1.3	177.9 ± 6.4	77.8 ± 8.0	Trained

n sample size

Table 2 Summary and characteristics of the studies included in this review

Study	Study design	Resistance exercise(s)	Exercise protocols	Velocity of movement	Training volume	Outcome measure(s)
Arent et al. [25]	Counterbalanced order and crossover design	Bench press Lat pulldown Shoulder press Seated rows Triceps extensions Biceps curls	RTF: 3 × 10 @ 100% 10RM RTNF 1: 3 × 10 @ 70% 10RM RTNF 2: 3 × 10 @ 40% 10RM 90 s interval between sets	Not reported	Not equalized	RPE (au)—ranged from 6 to 20
Fonseca et al. [36]	Controlled, randomized, and crossover design	Back squat	RTF: 4 × 11.9 ± 0.4 @ 12RM RTNF: 8 × 6 @ 12RM 180 s interval between sets	Not reported	Equalized	Vertical jump height—CMJ (cm) Power output for back squat (W) RPE (au)—ranged from 0 to 10
González-Badillo et al. [26]	Crossover design	Bench press Back squat	RTF: 3 × ~8RM (corresponding to @80% 1RM) RTNF: 3 × 4 (corresponding to @80% 1RM) 300 s interval between sets	Concentric phase in an explosive manner, at maximum intended velocity and eccentric phase not reported	Not equalized	Vertical jump height—CMJ (%) Velocity of movement against V ₁ -load for bench press (%) Velocity of movement against V ₁ -load for back squat (%) Muscle damage—CK (UI·L ⁻¹)
González-Hernández et al. [6]	Randomized and counterbalanced order and repeated measures crossover design	Back squat	RTF: 6 × 10 @ 10RM (~75% 1RM) RTNF: 6 × 5 @ 10RM (~75% 1RM) 300 s interval between sets	Concentric phase as fast as possible and eccentric phase in controlled self-selected velocity	Not equalized	Isometric strength—maximal voluntary isometric contraction for leg extension (N) Muscle damage—CK (UI·L ⁻¹)
Gorostiaga et al. [12]	Randomized order and crossover design	Leg press	RTF: 5 × 10 @ 10RM (79.1 ± 8.3% 1RM) RTNF: 10 × 5 @ 10RM (79.1 ± 8.3% 1RM) 120 s interval between sets	Displace the weight as fast as possible	Equalized	Power output for leg press (W) Metabolic response—lactate (mmol·L ⁻¹) Metabolic response—ammonia (μmol·L ⁻¹)
Gorostiaga et al. [27]	Randomized order and crossover design	Leg press	RTF: 5 × 10 @ 10RM (83% 1RM) RTNF: 10 × 5 @ 10RM (83% 1RM) 120 s interval between sets	Displace the weight as fast as possible	Equalized	Power output for leg press (W)
Gorostiaga et al. [34]	Randomized order and crossover design	Leg press	RTF: 1 × 10 @ 10RM (~85% 1RM) RTNF: 1 × 5 @ 10RM (~85% 1RM)	Displace the weight as fast as possible	Not equalized	Power output for leg press (W) Metabolic response—lactate (mmol·L ⁻¹) Metabolic response—ammonia (μmol·L ⁻¹)

Table 2 (continued)

Study	Study design	Resistance exercise(s)	Exercise protocols	Velocity of movement	Training volume	Outcome measure(s)
Hiscock et al. [52]	Randomized and crossover design	Bench press Leg press Lat pull down Leg curl Triceps pushdown	RTF 1: 3×MNR@70% 1RM, 60 s interval between sets RTF 2: 3×MNR @70% 1RM, 180 s interval between sets RTF 3: 3×MNR @40% 1RM, 60 s interval between sets RTF 4: 3×MNR @40% 1RM, 180 s interval between sets RTNF 1: 3×8 @70% 1RM, 180 s interval between sets RTNF 2: 3×14 @40% 1RM, 180 s interval between sets (matched tonnage with RTNF 1)	2 s concentric phase, 1 s pause, 2 s eccentric phase and 1 s pause	Not equalized	RPE (au)—ranged from 0 to 10
Linnamo et al. [29]	Non-randomized order and crossover design	Sit-up (no load) Bench press Leg extension (main exercise)	RTF: 5×10 @10RM RTNF: 5×10 @40% 10RM 120 s interval between sets	RTF: Concentric phase with own velocity and resisting the load during the eccentric phase RTNF: Perform each repetition as rapidly (explosively) as possible throughout each set	Not equalized	Isometric strength—maximal voluntary isometric contraction for leg extension (leg press) [%] Metabolic response—lactate (mmol·L ⁻¹)
Linnamo et al. [30]	Non-randomized order and crossover design	Sit-up (no load) Bench press Leg extension (main exercise)	RTF: 5×10 @10RM RTNF 1: 5×10 @40% 10RM RTNF 2: 5×10 @70% 10RM 120 s interval between sets	RTF and RTNF 2: Submaximal ‘steady’ velocities RTNF 1: Concentric and phase performed as fast as possible	Not equalized	Isometric strength—maximal voluntary isometric contraction for leg extension (leg press) [%] Metabolic response—lactate (mmol·L ⁻¹)
Martorelli et al. [49]	Counterbalanced order and crossover design	Back squat Bench press	RTF 1: 5×MNR @75% 1RM, 120 s interval between sets RTF 2: 5×MNR @90% 1RM, 180 s interval between sets RTNF: 5×6 @50% 1RM, 120 s interval between sets	2 s concentric phase and 2 s eccentric phase RTNF: concentric phase was performed as fast as possible	Not equalized	Metabolic response—lactate (mmol/L) Muscle damage—CK (U/L) RPE (au)—ranged from 0 to 10

Table 2 (continued)

Study	Study design	Resistance exercise(s)	Exercise protocols	Velocity of movement	Training volume	Outcome measure(s)
McGuigan et al. [53]	Randomized and crossover design	Back squat Bench press	RTF: 6 × 10RM @75% 1RM RTNF: 3 × 10 @30% 1RM 120 s interval between sets	Not reported	Not equalized	RPE (au)—ranged from 0 to 10
Morán-Navarro et al. [7]	Randomized, controlled, and crossover design	Bench press Back squat	RTF: 3 × 10 @75% 1RM RTNF 1: 3 × 5 @75% 1RM RTNF 2: 6 × 5 @75% 1RM 300 s interval between sets	Concentric phase in maximal intended velocity and eccentric phase not reported	Equalized in one of the conditions	Vertical jump height—CMJ (cm) Velocity of movement against V ₁ -load for bench press (m·s ⁻¹) Velocity of movement against V ₁ -load for back squat (m·s ⁻¹) Metabolic response—ammonia (μmol·L ⁻¹) Muscle damage—CK (IU·L ⁻¹)
Pareja-Blanco et al. [8]	Randomized order and crossover design	Bench press Back squat	RTF 1: 3 × ~12RM @70% estimated 1RM RTNF 1: 3 × 6 @70% estimated 1RM RTF 2: 3 × ~10RM @75% estimated 1RM RTNF 2: 3 × 5 @75% estimated 1RM RTF 3: 3 × ~8RM @80% estimated 1RM RTNF 3: 3 × 4 @80% estimated 1RM RTF 4: 3 × ~6RM @85% estimated 1RM RTNF 4: 3 × 3 @85% estimated 1RM RTF 5: 3 × ~4RM @90% estimated 1RM RTNF 5: 3 × 2 @90% estimated 1RM 300 s interval between sets	Concentric phase in maximal intended velocity and eccentric phase at a controlled pace (0.40–0.70 m·s ⁻¹)	Not equalized	Vertical jump height—CMJ (%) Velocity of movement against V ₁ -load for bench press (%) Velocity of movement against V ₁ -load for back squat (%) Muscle damage—CK (IU·L ⁻¹)

Table 2 (continued)

Study	Study design	Resistance exercise(s)	Exercise protocols	Velocity of movement	Training volume	Outcome measure(s)
Pareja-Blanco et al. [32]	Randomized order and crossover design	Bench press Back squat	RTF: $3 \times \sim 10\text{RM}$ @75% estimated 1RM RTNF 1 (pre-training period): 3×5 @75% estimated 1RM RTNF 2 (post 10-week training period): 3×5 @75% estimated 1RM 300 s interval between sets	Concentric phase in maximal intended velocity and eccentric phase at a controlled pace ($\sim 0.50 \text{ m}\cdot\text{s}^{-1}$)	Not equalized	Vertical jump height—CMJ (%) Velocity of movement against V_1 -load for bench press (%) Velocity of movement against V_1 -load for back squat (%) Muscle damage—CK ($\text{IU}\cdot\text{L}^{-1}$)
Pareja-Blanco et al. [58]	Randomized order and crossover design	Bench press Back squat	RTF: $3 \times \sim 12\text{RM}$ @70% 1RM RTNF: 3×6 @70% 1RM 300 s interval between sets	Concentric phase in maximal intended velocity and eccentric phase not reported	Not equalized	Vertical jump height—CMJ (cm) Velocity of movement against V_1 -load for bench press (%) Velocity of movement against V_1 -load for back squat (%) Muscle damage—CK ($\text{IU}\cdot\text{L}^{-1}$)
Raastad et al. [33]	Counterbalanced order and crossover design	Back squat Front squat Leg extension	RTF: 3×3 @3RM for squats, 360 s interval between sets 3×6 @6RM for leg extension, 240 s interval between sets RTNF: 3×3 @70% 3RM for squats, 360 s interval between sets 3×6 @76% 6RM for leg extension, 240 s interval between sets	Not reported	Not equalized	Metabolic response—lactate ($\text{mmol}\cdot\text{L}^{-1}$)

Table 2 (continued)

Study	Study design	Resistance exercise(s)	Exercise protocols	Velocity of movement	Training volume	Outcome measure(s)
Sánchez-Medina and González-Badillo et al. [9]	Parallel design for the exercises and crossover for the conditions design	Bench press or back squat	RTF 1: 3 × 12 @ 12RM (~70% 1RM) RTNF 1: 3 × 6 @ 12RM (~70% 1RM) RTNF 2: 3 × 8 @ 12RM (~70% 1RM) RTNF 3: 3 × 10 @ 12RM (~70% 1RM) RTF 2: 3 × 10 @ 10RM (~75% 1RM) RTNF 4: 3 × 6 @ 10RM (~75% 1RM) RTNF 5: 3 × 8 @ 10RM (~75% 1RM) RTF 3: 3 × 8 @ 8RM (~80% 1RM) RTNF 6: 3 × 4 @ 8RM (~80% 1RM) RTNF 7: 3 × 6 @ 8RM (~80% 1RM) RTF 4: 3 × 6 @ 6RM (~85% 1RM) RTNF 8: 3 × 3 @ 6RM (~85% 1RM) RTNF 9: 3 × 4 @ 6RM (~85% 1RM) RTF 5: 3 × 4 @ 4RM (~90% 1RM) RTNF 10: 3 × 2 @ 4RM (~90% 1RM) 300 s interval between sets	Concentric phase in an explosive manner, at maximal intended velocity and eccentric phase in velocity controlled	Not equalized	Vertical jump height—CMJ (%) Velocity of movement against V ₁ -load for bench press (%) Velocity of movement against V ₁ -load for back squat (%) Metabolic response—ammonia (μmol·L ⁻¹) Metabolic response—lactate (mmol·L ⁻¹)
Shibata et al. [31]	Non-randomized order and crossover design	Back squat	RTF: 3 × MNR @ 75% 1RM RTNF: 6 × total repetitions performed in RTF @ 75% 1RM 180 s interval between sets	2 s concentric phase and 2 s eccentric phase	Equalized	Isometric strength—maximal voluntary isometric contraction for leg extension (N) Metabolic response—lactate (mmol/L) RPE (au)—ranged from 0 to 10

Table 2 (continued)

Study	Study design	Resistance exercise(s)	Exercise protocols	Velocity of movement	Training volume	Outcome measure(s)
Vasquez et al. [28]	Randomized order and crossover design	Back squat	RTF 1: 1 × MNR @ 50% 1RM RTNF 1: 1 × 3 @ 50% 1RM RTF 2: 1 × MNR @ 70% 1RM RTNF 2: 1 × 3 @ 70% 1RM RTF 3: 1 × MNR @ 90% 1RM RTNF 3: 1 × 3 @ 90% 1RM 600 s interval between conditions	Concentric phase as fast as possible eccentric phase in velocity controlled	Not equalized	Power output for back squat (W/kg) RPE (au)—ranged from 6 to 20

au arbitrary units, CK creatine kinase, CMJ countermovement jump, MNR maximum number of repetitions, RPE rating of perceived exertion, RM repetition maximum, RTF resistance training performed to failure, RTNF resistance training not performed to failure, @ load used

showed that RPE was greater in RTF than RTNF when the volume was not equalized (SMD 2.04, 95% CI 0.80–3.28, $p=0.001$), although high inconsistency and significant heterogeneity were found in the overall analysis ($I^2=83.72\%$, $p<0.01$). Figure 4 summarizes all types of fatigue responses on RTF and RTNF over time.

3.5 Publication Bias and Certainty of Evidence

Regarding publication bias, it was not possible to perform an adequate analysis because no analysis included 10 or more studies (see ESM). Table 4 shows details of the certainty of evidence (GRADE). There was low certainty of evidence for biomechanical properties, metabolic response, and RPE meta-analyses (GRADE score=2) and moderate certainty of evidence for muscle damage meta-analysis (GRADE score=3).

4 Discussion

This systematic review and meta-analysis compared RTF versus RTNF on acute fatigue. Among the main findings, there was a significant reduction in biomechanical properties and a greater increase in metabolic response, muscle damage (mainly after 48 h), and RPE following RTF compared with RTNF.

4.1 Biomechanical Properties

There was a large reduction in biomechanical properties with RTF compared with RTNF (SMD -0.96). Different time under tension, which refers to the period in which the muscle undergoes external loading during a set, by dynamic or isometric contractions [54], may in part explain this result. Although the studies included in this meta-analysis did not report an equalization of time under tension between the RTF and RTNF protocols [6–9, 29, 31, 32, 36], the higher number of repetitions performed in RTF, and particularly the higher number of concentric movements, are determinant factors for increasing time under tension and, consequently, muscle fatigue [54]. A prolonged time under tension associated with muscle failure led to an increase in blood lactate and blood ammonia concentration and may explain at least part of the peripheral fatigue [12, 27, 34, 55]. The increase in blood metabolic markers is a result of increased use of lactic glycolysis as a source of energy, which is associated with reduced intramuscular ATP, and thus leads to a decrease in maximal strength and power output [12, 27, 34, 35, 56].

A complex interaction between central and peripheral fatigue mechanisms influences the muscle potential to perform muscle actions [55]. A recent study showed no difference in central fatigue between RTF and RTNF when a

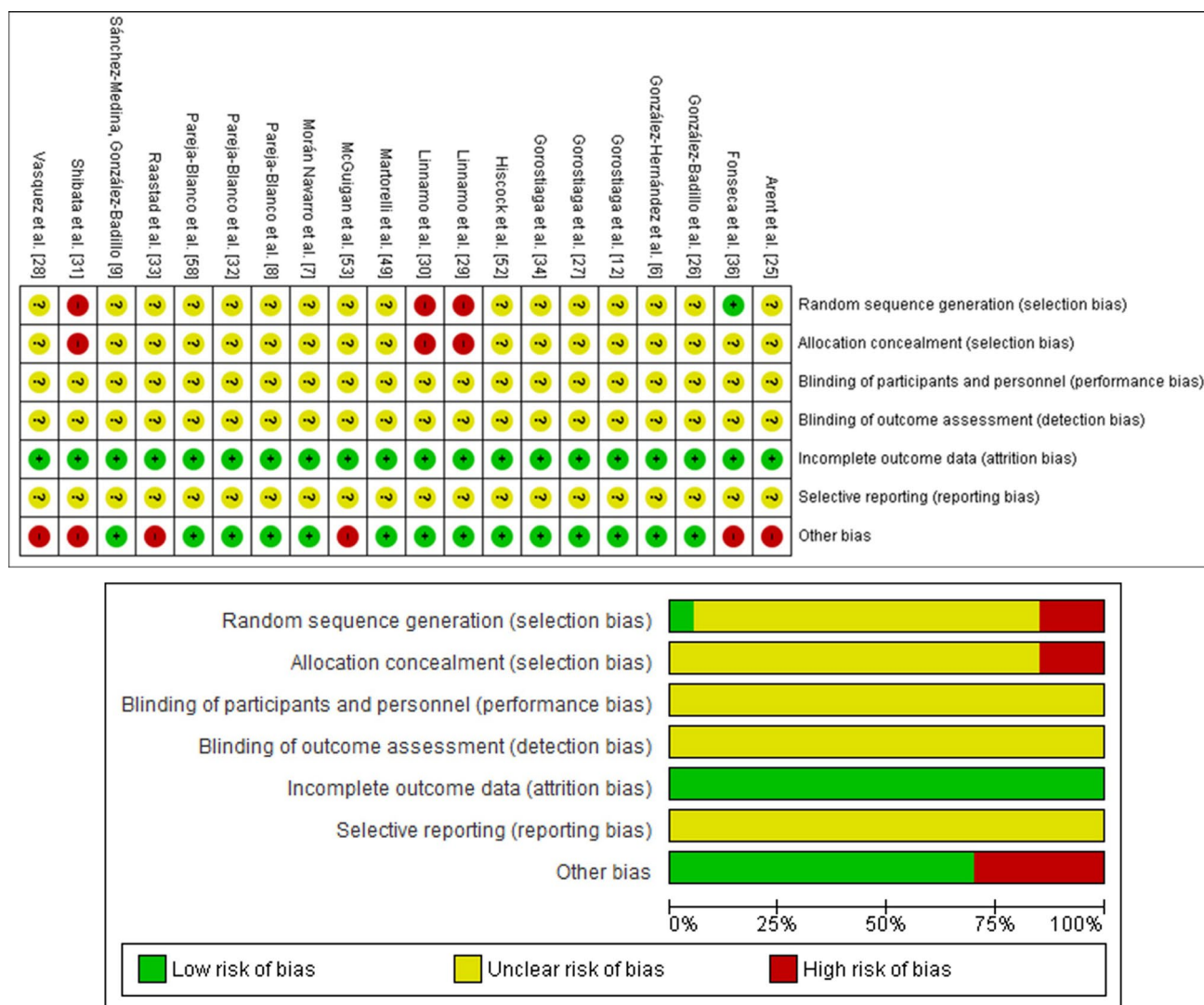


Fig. 2 Risk of bias in the included studies

full-squat exercise was performed with a 10RM load [6]. On the other hand, previous studies have shown a correlation between increased peripheral fatigue and decreased performance [12, 27, 34]. RTF may lead to a higher reduction in skeletal muscle $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity, sarcoplasmic reticulum Ca^{2+} release, and Ca^{2+} uptake [57]; thus, a higher reduction in RTF biomechanical properties could be explained by the increase in peripheral fatigue [6, 12, 27].

It is noteworthy that no other factors besides the type of tests (CMJ, MVIC, velocity BP, and velocity SQ) explained the true heterogeneity between studies for RTF effects on biomechanical properties. There was a significantly greater reduction in RTF biomechanical properties than RTNF when the velocity BP test was used [7–9, 32] compared with the tests assessing biomechanical properties in lower limbs [6–9, 29, 31, 32, 36]. The larger reduction in performance of the upper-limb muscles may reflect the higher rate of decrease

in relative average velocity [26, 58]. Izquierdo et al. [59] observed that the decline in the velocity of movement in the bench press occurred when the number of repetitions exceeded 34% of the total performed; however, velocity of movement declined in the squat when only 48% of repetitions were exceeded. In addition, the velocity of movement achieved during the last repetition was significantly higher in the squat than in the bench press. Other possible factors leading to different degrees of fatigue between upper- and lower-limb muscles may be associated with different maximal strength levels, muscle fiber distribution, muscle size [60, 61], muscle mechanics (i.e., length and muscle pennation angle), and the functional differences of the muscles [62].

The present meta-analytic result for biomechanical properties may provide important insights regarding long-term adaptations. In general, it appears that large acute fatigue

Fig. 3 Forest plot of standardized mean difference [biomechanical properties (a), muscle damage (c), and RPE (d)] and raw mean difference [metabolic response (b)] for RTF and RTNF. *CI* confidence interval, *df* degrees of freedom, *I*² inconsistency between studies, *RMD* raw mean difference, *RPE* rating of perceived exertion, *RTF* resistance training performed to failure, *RTNF* resistance training not performed to failure, *SMD* standardized mean difference

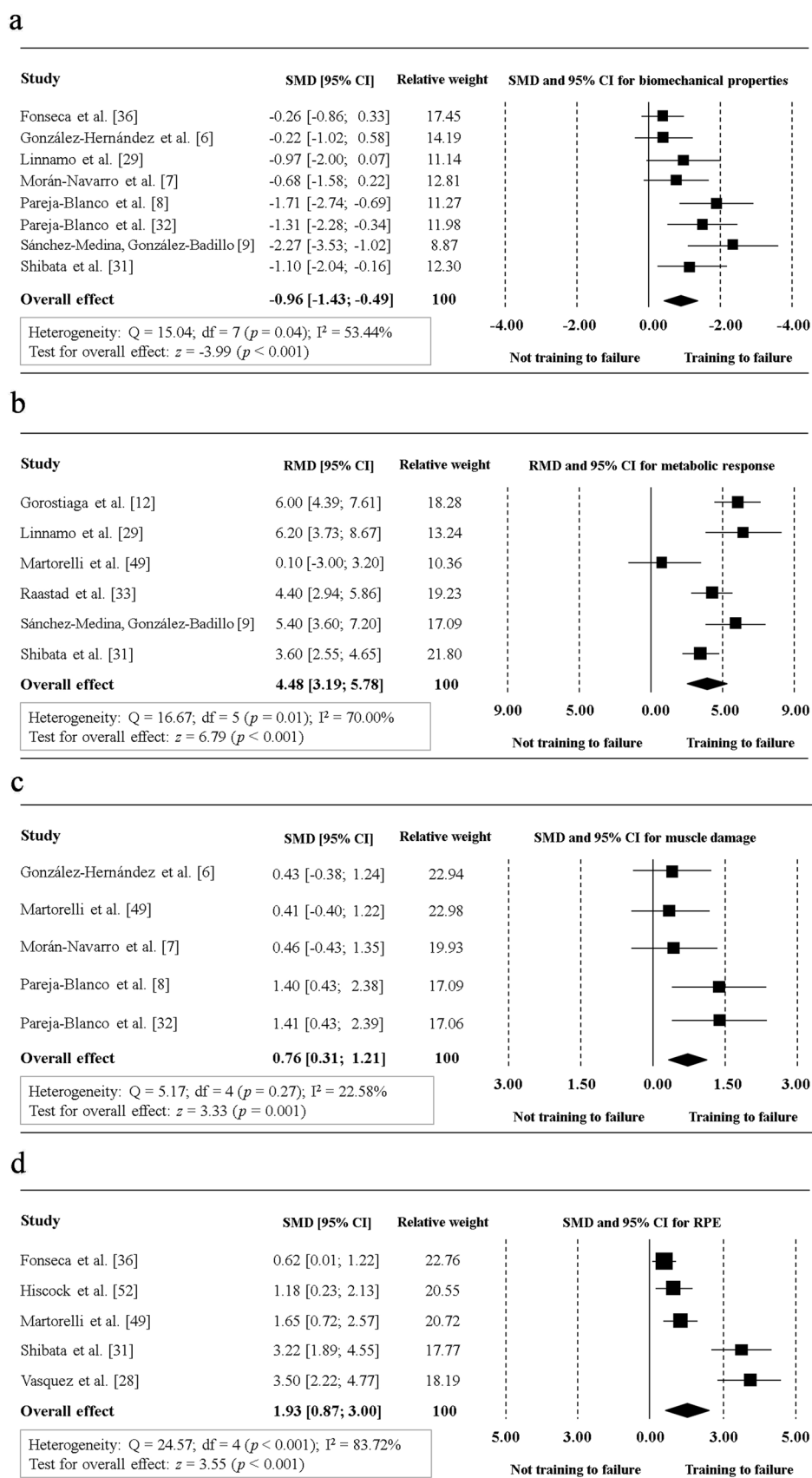


Table 3 Results of the subgroup analyses

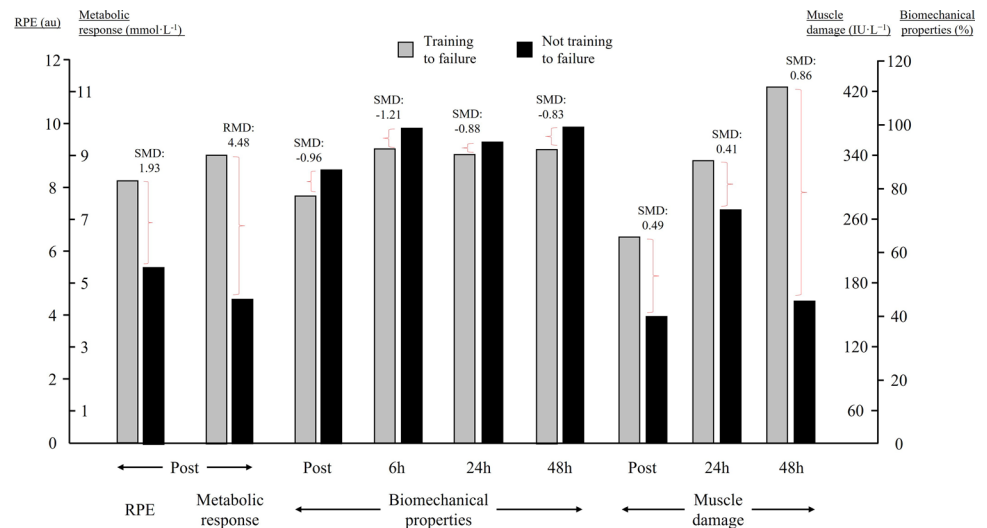
Independent or dependent variables	SMD or RMD	95% CI	<i>p</i> -value	<i>I</i> ² (<i>p</i> -value)	<i>k</i>	<i>p</i> -value (difference)
Biomechanical properties (SMD)						
Training status						
Trained	−0.93	−1.53 to −0.33	0.002	63.79 (0.02)	6	0.92
Untrained	−0.88	−1.57 to −0.20	0.01	0.00 (0.83)	2	–
Type of test						
CMJ (cm or %)	−1.14	−1.86 to −0.43	0.002	67.54 (0.02)	5	<0.001
MVIC (<i>N</i> or %)	−0.70	−1.26 to −0.13	0.02	13.06 (0.32)	3	–
Velocity BP (m·s ^{−1} or %)	−2.74	−3.35 to −2.13	<0.001	0.00 (0.97)	4	–
Velocity SQ (m·s ^{−1} or %)	−1.65	−2.24 to −1.06	<0.001	20.39 (0.29)	4	–
Time point						
Immediately post-exercise (until 60 s)	−0.96	−1.43 to −0.49	<0.001	53.44 (0.04)	8	0.89
6 h post-exercise	−1.21	−2.01 to −0.41	0.003	51.06 (0.13)	3	–
24 h post-exercise	−0.88	−1.49 to −0.27	0.005	60.04 (0.03)	6	–
48 h post-exercise	−0.83	−1.39 to −0.26	0.004	45.73 (0.12)	5	–
Load						
70%	−0.81	−1.28 to −0.33	0.001	74.49 (0.05)	2	0.10
75%	−1.10	−2.04 to −0.16	0.02	51.32 (0.07)	6	–
80%	−3.10	−4.55 to −1.65	<0.001	0.00 (0.37)	2	–
85%	−2.82	−4.21 to −1.44	<0.001	80.40 (0.02)	2	–
90%	−1.90	−3.09 to −0.72	0.002	31.01 (0.23)	2	–
Volume						
Equalized	−0.56	−1.04 to −0.09	0.02	11.74 (0.32)	3	0.12
Not equalized	−1.22	−1.91 to −0.53	0.001	58.16 (0.05)	5	–
Metabolic response (RMD)						
Sensitivity analyses						
Ammonia (μmol·L ^{−1})	44.66	32.27–57.05	<0.001	56.41 (0.10)	3	–
Volume						
Equalized	4.72	2.37–7.07	<0.001	83.31 (0.01)	2	0.77
Not equalized	4.46	2.95–5.96	<0.001	71.58 (0.01)	4	–
Muscle damage (SMD)						
Time point						
Immediately post-exercise (until 5 min) [IU·L ^{−1}]	0.49	−0.03 to 1.00	0.06	0.00 (0.92)	3	0.42
24 h post-exercise (IU·L ^{−1})	0.41	−0.07 to 0.89	0.09	0.00 (0.94)	3	–
48 h post-exercise (IU·L ^{−1})	0.86	0.33–1.42	0.002	29.75 (0.23)	4	–
RPE (SMD)						
Volume						
Equalized	1.85	−0.70 to 4.40	0.16	91.83 (<0.001)	2	0.90
Not equalized	2.04	0.80–3.28	0.001	76.60 (0.01)	3	–

CI confidence interval, *CMJ* countermovement jump, *I*² inconsistency between studies, *k* number of studies, *MVIC* maximal voluntary isometric contraction, *RPE* rating of perceived exertion, *RMD* raw mean difference, *SMD* standardized mean difference, *Velocity BP* velocity of movement against V₁-load for bench press, *Velocity SQ* velocity of movement against V₁-load for squat

after RTF does not seem to be optimal for inducing greater strength, hypertrophy, and power gains [15–18]. It is not clear whether the different body segments exercised may affect the difference between RTF and RTNF adaptations [17, 18], however it is tempting to speculate that the greater

muscle hypertrophy in upper limbs following RTF [18] was stimulated by higher fatigue response within this body segment. Additionally, it is important to mention that the relationship between acute fatigue and muscle hypertrophy is still inconclusive.

Fig. 4 Changes in biomechanical properties, metabolic response, muscle damage, and RPE for RTF and RTNF. *au* arbitrary units, *RMD* raw mean difference, *RPE* rating of perceived exertion, *RTF* resistance training performed to failure, *RTNF* resistance training not performed to failure, *SMD* standardized mean difference



4.2 Metabolic Response

There was a large increase in metabolic response, assessed by lactate concentration in RTF compared with RTNF (RMD 4.48 mmol·L⁻¹). A possible explanation for this result may be that RTF demands maximal effort [28, 31, 53] that leads to higher increases in muscle lactate levels [27]. A positive relationship between muscle lactate and blood lactate ($r=0.90$) [12] explains the higher blood lactate concentration after RTF compared with RTNF found in our meta-analysis. **Acidification of the internal environment is a process arising from the increase in hydrogen ions and, concomitantly, a reduction in blood pH [63, 64] that reduce muscular functions by the following mechanisms: (1) reduction in the transition from the cross-bridge from the low to the high strength state; (2) inhibition of sarcomere shortening velocity; (3) inhibition of myofibrillar ATPase; (4) inhibition of the glycolytic rate; (5) reduction in cross-bridge activation by competitively inhibiting Ca²⁺ binding to troponin C; and (6) reduction in Ca²⁺ uptake by inhibiting sarcoplasmic ATPase (leading to a subsequent reduction in Ca²⁺ release) [63, 64].** Gorostiaga et al. [12] observed that peak power output during the leg press begins to decline after the power generated during the first two repetitions (100%) when the blood lactate concentration exceeds ~5–6 mmol·L⁻¹. Thus, considering the average values of our results (i.e., 9.8 ± 2.2 mmol·L⁻¹ in RTF compared with 4.0 ± 1.3 mmol·L⁻¹ in RTNF), it seems evident that neuromuscular performance has declined more in RTF.

A larger increase in metabolic response after RTF compared with RTNF was also shown within studies assessing ammonia concentration (RMD 44.66 μmol·L⁻¹). Indeed, it has been reported that RTF leads to a higher concentration of blood metabolites and a considerable depletion of muscle purines, while RTNF allows the cellular maintenance

of homeostasis [9, 12, 27]. During high-intensity exercise and RTF, the reduced ADP rephosphorylation capacity and high ATP turnover rate may result in a reduced concentration of muscular ATP and an approximately stoichiometric increase in adenosine monophosphate deamination to inosine monophosphate and ammonia [12, 65]. Similar to lactate response, the ammonia concentration may explain the decline of biomechanical properties, since the ~40 μmol·L⁻¹ elevation of blood ammonia concentration is associated with the decline in peak power ($r=-0.87$) [12]. In addition, the relationship between metabolic response and biomechanical properties is related to the type of exercise performed. Blood ammonia begins to increase constantly above the resting values only when there is a considerable loss in the velocity of movement (over ~30% for squat exercise and ~35% for the bench press exercise) [9].

4.3 Muscle Damage

RTF led to higher muscle damage compared with RTNF (SMD 0.76). Despite the fact that both RTF and RTNF seem to increase muscle damage immediately and 24 h post the end of the training session, no difference was found between RTF and RTNF in these time points. The significantly higher muscle damage was found at 48 h after RTF compared with RTNF and seems to be related to the fact that at this time point, the level of muscle damage induced by RTNF has already begun to decrease. In theory, the higher number of eccentric and concentric actions (number of repetitions) performed during RTF compared with RTNF is known to increase muscle damage [66]. In fact, higher muscle damage occurs with a greater number of repetitions independently of being performed to failure [7, 8]. **Even when the volume was equalized for RTF and RTNF, greater muscle damage was seen for RTF [7], reinforcing that the achievement of**

Table 4 Summary of meta-analysis findings and certainty of evidence

Outcomes	Comparison	Summary of findings		Certainty of evidence (GRADE)							
		<i>k</i>	<i>n</i>	Effect (95% CI)	Effect direction	Risk of bias	Inconsistency	Indirectness	Imprecision	Publication bias	Certainty
Biomechanical properties	RTF×RTNF	8	90	−0.96 (−1.43 to −0.49)	↓	Not serious	Serious	Not serious	Serious	Undetected	⊕⊕○○ Low
		6	60	4.48 (3.19–5.78)	↑	Not serious	Serious	Not serious	Serious	Undetected	⊕⊕○○ Low
Metabolic response	RTF×RTNF	5	54	0.76 (0.31–1.21)	↑	Not serious	Not serious	Not serious	Serious	Undetected	⊕⊕⊕○ Moderate
Muscle damage	RTF×RTNF	5	66	1.93 (0.87–3.00)	↑	Not serious	Serious	Not serious	Serious	Undetected	⊕⊕○○ Low
RPE	RTF×RTNF	5	66	1.93 (0.87–3.00)	↑	Not serious	Serious	Not serious	Serious	Undetected	⊕⊕○○ Low
CI confidence interval, <i>k</i> number of studies, <i>n</i> sample size, <i>RPE</i> rating of perceived exertion, <i>RTF</i> resistance training performed to failure, <i>RTNF</i> resistance training not performed to failure											

CI confidence interval, *k* number of studies, *n* sample size, *RPE* rating of perceived exertion, *RTF* resistance training performed to failure, *RTNF* resistance training not performed to failure

repetitions closer to failure may be a critical factor for inducing muscle damage.

Some confounding factors, such as interindividual variability on creatine kinase levels [50, 67] and the repeated bout effect [68], could mask pronounced muscle damage, which in turn may cause severe health consequences [69–71]. Hence, other muscle damage markers, such as muscle pain, transient loss of muscle strength, and local inflammation, should be considered in a comprehensive muscle damage assessment. Despite this, creatine kinase is still among the more frequently used measures of muscle damage.

It is noteworthy that muscle damage does not necessarily lead to elevated blood creatine kinase levels. For example, Çakir-Atabek et al. [72] showed that after 60 maximal voluntary eccentric contractions of the elbow flexors, the decline in strength was associated with a higher oxidative stress environment but was not related to blood creatine kinase levels. Specifically, this oxidative environment was characterized by an increase in oxidation of carbonylated protein, increased total oxidative state, and higher oxidative stress index (percentage ratio of oxidant status to antioxidant status). Meanwhile, since the increase in CK is higher 48 h after RTF than RTNF, it is likely that overall muscle damage in RTF was higher or more persistent than in RTNF.

4.4 Rating of Perceived Exertion

Higher RPE scores were found after RTF when compared with RTNF (SMD 1.93). The higher RPE scores for RTF could be caused by general fatigue, muscle fatigue, cardiovascular stress, and discomfort [73]. Specifically, the other outcomes assessed in the present study have been associated with RPE in previous studies, such as higher lactate concentration [74] and decreased power output [75]. Nevertheless, there is an important association between mental fatigue and exercise tolerance [76], and the higher anxiety and tiredness observed in RTF sessions could be the cause of higher RPE compared with RTNF [25].

Different variables regulate RPE scores after RT. Our results showed that training volume may influence the RPE after RTF, however such influence was not observed after RTNF; such differences can be related to training volume. We found high heterogeneity and a large difference in ES in the RPE score for equalized volume [31, 36]. Other studies have shown that muscle failure leads to higher RPE scores independently of load and training volume [49, 77]. Fisher et al. [77] demonstrated that independent of the load used (high or low load), all participants reported maximal RPE scores in each set and exercise session, however the degrees of discomfort were different between the intensities, suggesting that maximal effort required in the sets determined the RPE.

In summary, the current meta-analytic result for RPE may be due to higher fatigue and mental stress, but mainly due to maximal effort required in RTF. The fact that studies relate the maximum score of RPE to the inability to perform an exercise with maximum exertion [77] and the characteristics of the RTNF protocols designed to prevent muscle failure in the studies included in our analysis, was likely the main contributor of the large SMD identified.

4.5 Future Research

While the total number of studies included in this meta-analysis was relatively high, there is a lack of research including highly trained participants, who would be the most likely candidates for advanced resistance-training methods, which require future study. Future research should also compare RTF with the lowest number of reserve repetitions in the RTNF protocols and equalize the movement tempo used [78, 79] and with controlled time under tension during the RT protocols [54], to better isolate the effects arising the failure stimulus itself. Since only one study described the course of the acute fatigue up to 72 h [7] and none of the studies assessed the results for longer periods, there is a need for more clarification regarding the duration of RTF effects on the human body. Furthermore, considering RTF protocols may require longer time intervals between sessions (lower frequency) for complete recovery compared with RTNF protocols, future studies should investigate whether these frequency-adjusted protocols may induce different neural and morphological long-term responses.

4.6 Limitations

The present study combined data from different studies, aiming to compare RTF versus RTNF on acute fatigue markers. These studies did not define muscle failure using the same criteria and compared protocols with different numbers of repetitions, which could have affected the overall effects found.

The moderate to high inconsistency between studies (biomechanical properties [$I^2 = 53.44\%$], metabolic response [$I^2 = 70.00\%$], and RPE [$I^2 = 83.72\%$]) suggest some considerable difference among studies, however subgroup analyses were performed to isolate those differences and clarify at least part of the cause of different effects.

In the present systematic review, there were multiple publications based on data of the same participants [26, 27, 30, 34, 58]. To counteract the bias caused by this sample overlapping, a very common mistake in many meta-analyses, we did not repeat the results from the same individuals in the main analyses, by giving preference to one main trial within each study (according to the type of assessment of the outcome, load of the protocol, and time point assessed).

Furthermore, all the variations presented within studies that could be comparable were discriminated in secondary subgroup analyses.

5 Conclusions

The results of the present systematic review and meta-analysis showed that RTF caused a greater increase in all the acute fatigue markers in healthy adult males when compared with RTNF, which includes the decline in biomechanical properties, increase in metabolic response, muscle damage, and RPE. Although the absence of training volume equalization was an important limitation of some studies included in the meta-analysis, an exploratory analysis of subgroups did not show any influence of training volume (regardless of equalization) on the results. Our findings allow a comprehensive understanding about the time-course of acute fatigue, which is fundamental for proper RT prescription. The effects of exacerbated fatigue responses and the suggestive need for a longer interval between RT sessions for complete recovery require further attention to appropriate RT design. In particular, the frequency of RTF must be taken into account, considering that incomplete recovery would increase the risk for injury, overtraining, and psychological burnout, which could also impair long-term training adaptations.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s40279-021-01602-x>.

Declarations

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Conflicts of interest João Guilherme Vieira, Amanda Veiga Sardeli, Marcelo Ricardo Dias, José Elias Filho, Yuri de Almeida Campos, Leandro de Oliveira Sant'Ana, Luis Leitão, Victor Machado Reis, Michal Wilk, Jefferson da Silva Novaes, and Jeferson Macedo Vianna declare that they have no conflicts of interest relevant to the content of this systematic review and meta-analysis.

Availability of data and material The database that supports the conclusions of this systematic review is available from the corresponding author upon request.

Ethics approval Not applicable.

Consent Not applicable.

Author contributions João Guilherme Vieira, Amanda Veiga Sardeli, Marcelo Ricardo Dias, Jefferson Novaes, and Jeferson Vianna designed the manuscript, and João Guilherme Vieira wrote the first draft. João Guilherme Vieira and Marcelo Ricardo Dias conducted the literature search. João Guilherme Vieira and Marcelo Ricardo Dias wrote the Methods and Results, and prepared the tables/figures. Amanda Veiga Sardeli conducted the meta-analysis. João Guilherme Vieira and José Elias Filho reviewed the Methods and helped choose the tool to assess the quality of the articles used in this systematic review. Amanda Veiga Sardeli, José Elias Filho, Jefferson Novaes, and Jeferson Vianna systematically guided João Guilherme Vieira during the article writing process. Yuri Campos, Leandro Sant'Ana, Luis Leitão, Victor Reis, and Michal Wilk reviewed the manuscript and the English language and contributed technically to the quality of the manuscript. All authors read and approved the final manuscript.

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